

Image Defect Detection and Localization Using Anomaly Detection

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Abstract

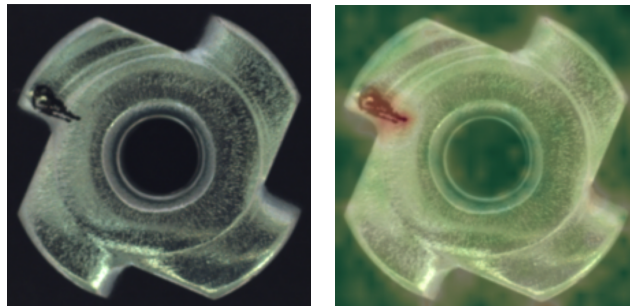
This work investigates the use of Machine Learning techniques for the automatic detection and localization of defects in images of industrial products. The proposed system uses a pre-trained ResNet-50 network for feature extraction, combined with the PatchCore method for anomaly detection. The approach is built using only defect-free images and is subsequently evaluated on both normal and anomalous samples. The objective is to correctly identify defective images and, through the generation of heatmaps, localize the regions most strongly associated with the presence of anomalies. The robustness of the system is further evaluated under domain shift, simulating realistic industrial acquisition variations such as illumination changes, sensor noise, and camera viewpoint shifts. The performance of the system is evaluated using standard binary classification metrics. On the metal_nut test set, the final configuration achieved an Accuracy of 91.3%, a Recall of 92.5%, and an F1-score of 94.5%.

1. Introduction

The automatic detection of defects in images represents one of the most relevant applications of computer vision in the industrial field. In quality control processes, manual product inspection can be slow, subjective, and difficult to scale, especially when a large number of samples must be analyzed. The use of Machine Learning techniques makes it possible to automate part of this process, supporting the identification of anomalous samples in a faster and more systematic way.

In this work, the problem of *anomaly detection* applied to images of industrial products is addressed. The objective is to distinguish normal samples from defective ones and, when possible, to localize the image regions associated with the presence of a defect. Unlike a traditional supervised approach, the system is trained using only non-defective images. In this way, the model learns a representation of the normal appearance of the product and uses this knowledge to identify deviations in test samples.

The developed approach is based on the use of a pre-



(a) Original image.

(b) Superimposed heatmap.

Figure 1: Example of defect detection and localization through anomaly detection.

trained ResNet-50[3] convolutional neural network as a feature extractor, combined with PatchCore[4], an anomaly detection method based on the comparison of local features. The representations extracted from intermediate layers of the network are used to build a *memory bank* containing the description of normal images only. During the inference phase, the features extracted from a new image are compared with those stored in the memory bank: a large distance from the normal samples indicates the presence of an anomaly.

In addition to producing a global anomaly score for each image, the system is able to generate a *heatmap*, which is useful for highlighting the areas most strongly associated with the defect. An example of the obtained result is shown in Fig. 1, where the original image and the corresponding visualization with the superimposed heatmap are compared.

The performance of the system is evaluated using standard binary classification metrics, including Accuracy, Precision, Recall, and F1-score. The qualitative analysis of the heatmaps also makes it possible to assess the ability of the model not only to distinguish normal and defective images, but also to approximately localize anomalous regions.

Finally, the robustness of the system is analyzed in the presence of variations in the test images, introducing alterations such as noise, blur, illumination changes, and geometric transformations. The objective is to evaluate the behavior of the model under conditions that differ from the

nominal ones and to analyze the impact of *domain shift* on the performance of the system.

2. Dataset

For the development and evaluation of the proposed system, the MVTec Anomaly Detection (MVTec AD)[1] dataset was used. The dataset is designed for industrial anomaly detection and contains high-resolution images belonging to different categories of objects and materials. For each category, both defect-free samples and images affected by different types of anomalies are provided.

In this work, the analysis was focused on the *metal_nut* category, which contains images of metallic nuts. Working on a single category allows the model to learn a more specific representation of the normal appearance of the object, avoiding the additional complexity introduced by combining products with very different visual characteristics.

The dataset follows the typical anomaly detection setting: the training set contains only images belonging to the *good* class, namely samples without defects. The test set instead includes both normal images and images affected by different types of anomalies. In this way, the system learns only from nominal samples and is later evaluated on its ability to detect deviations from the learned normal appearance.

Some examples from the *metal_nut* category are shown in Fig. 2. The first image represents a normal sample, while the following ones show different types of defects.

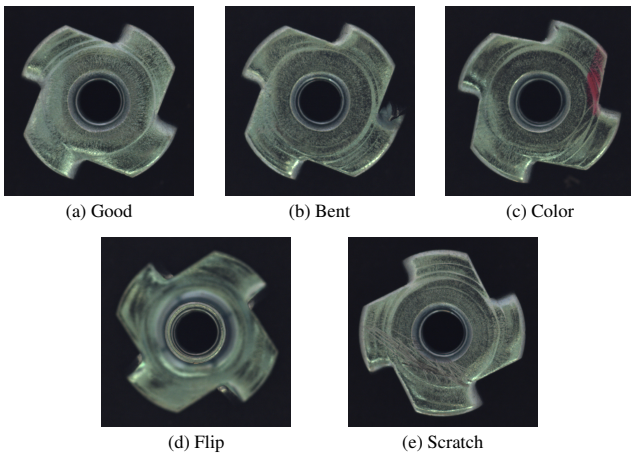


Figure 2: Examples from the *metal_nut* category of the MVTec AD dataset. Sample (a) represents a normal image, while (b)–(e) show different types of anomalies.

2.1. Preprocessing

Before being processed by the ResNet-50 network, all images undergo the same preprocessing pipeline. This step is required to obtain inputs with a consistent size and distribution, compatible with the pre-trained feature extractor.

Each image is first resized to 256 pixels and then center-cropped to a spatial resolution of 224×224 pixels. The resulting image is converted into a tensor and normalized using the mean and standard deviation values adopted by the pre-trained ResNet-50 model.

The normalization uses the standard ImageNet statistics, $\mu = [0.485, 0.456, 0.406]$ and $\sigma = [0.229, 0.224, 0.225]$, for the RGB channels.

The same preprocessing pipeline is applied to both training and test images. This ensures that the features stored in the memory bank and those extracted during inference are generated under the same input conditions.

Table 1: Preprocessing pipeline applied to each input image.

Operation	Parameters
Resize	256 pixels
Center Crop	224×224
Tensor Conversion	RGB image to tensor
Normalization	ImageNet statistics

2.2. Dataset Organization

The images are organized into training, validation, and test sets. The original MVTec AD training set for the *metal_nut* category contains 220 defect-free images. These samples are randomly divided into two subsets: 80% of the images are used to construct the normal memory bank, while the remaining 20% are used as a validation set for threshold selection.

Using a fixed random seed, this results in 176 training images and 44 validation images. Both subsets contain exclusively defect-free samples.

The test set contains both *good* samples and images affected by different anomaly types. During evaluation, all defective categories are considered anomalous, resulting in a binary classification problem between normal and anomalous samples.

The validation images are never included in the memory bank, ensuring that the anomaly threshold is estimated using normal samples that were not used to construct the reference feature representation.

Table 2: Number of images used for training, validation, and testing.

Subset / Category	Images
Good (Training)	176
Good (Validation)	44
Good (Test)	22
Bent	25
Color	22
Flip	23
Scratch	23
Total	335

Table 2 reports the number of images available for each category. The normal samples are separated between training and test sets, while the anomalous test samples are divided according to their specific defect type.

3. Methodology

The proposed anomaly detection system is based on the combination of a pre-trained ResNet-50 convolutional neural network and a PatchCore-based feature comparison approach. The overall pipeline can be divided into two main stages: the construction of a representation of normal samples and the evaluation of new test images.

During the first stage, only defect-free training images are processed by ResNet-50. Intermediate feature maps are extracted from the network and transformed into local feature vectors, representing different spatial regions of the input images. These vectors are collected into a *memory bank*, which represents the normal appearance of the analyzed object.

During inference, the same feature extraction procedure is applied to a test image. Each local feature vector is compared with the vectors stored in the memory bank. The distance from the most similar normal feature is used to estimate the degree of anomaly of each image region. These local distances are then used both to compute a global anomaly score and to generate a spatial anomaly map.

3.1. Feature Extraction with ResNet-50

A ResNet-50 model pre-trained on ImageNet[2] is used as the feature extractor. Since the objective is not to perform conventional image classification, the final classification output of the network is discarded. Instead, intermediate representations from *layer2* and *layer3* are collected through forward hooks.

The parameters of ResNet-50 are frozen during the entire procedure, preventing their modification and allowing the network to be used exclusively as a feature extractor. This also avoids the need to train a deep neural network from scratch using the relatively limited number of normal samples available for the selected object category.

The two selected layers provide feature maps at different levels of abstraction and spatial resolution. Features extracted from *layer2* retain more spatial information, while those obtained from *layer3* provide a higher-level representation of the image content.

Since the feature maps produced by the two layers have different spatial dimensions, the output of *layer3* is resized through bilinear interpolation to match the spatial resolution of *layer2*. The resulting tensors are then concatenated along the channel dimension, combining information from both levels of the network.

3.2. Memory Bank Construction

For each normal training image, the combined feature tensor has dimensions

$$B \times C \times H \times W, \quad (1)$$

where B represents the batch size, C the number of feature channels, and H and W the spatial dimensions of the feature map.

The tensor is rearranged so that each spatial location is represented by an independent feature vector of dimension C . Therefore, each image contributes $H \times W$ local descriptors to the memory bank.

The complete memory bank can be represented as

$$\mathcal{M} = \{m_1, m_2, \dots, m_N\}, \quad (2)$$

where each $m_i \in \mathbb{R}^C$ represents a local feature extracted from a defect-free training image.

The memory bank is constructed exclusively from the 176 normal images belonging to the training subset. The 44 validation images are not included in this process and are used only for threshold estimation.

Since the complete set of extracted patches is large, random subsampling is applied to reduce memory requirements and computational cost. In the final configuration, 20% of the available patch descriptors are retained in the memory bank.

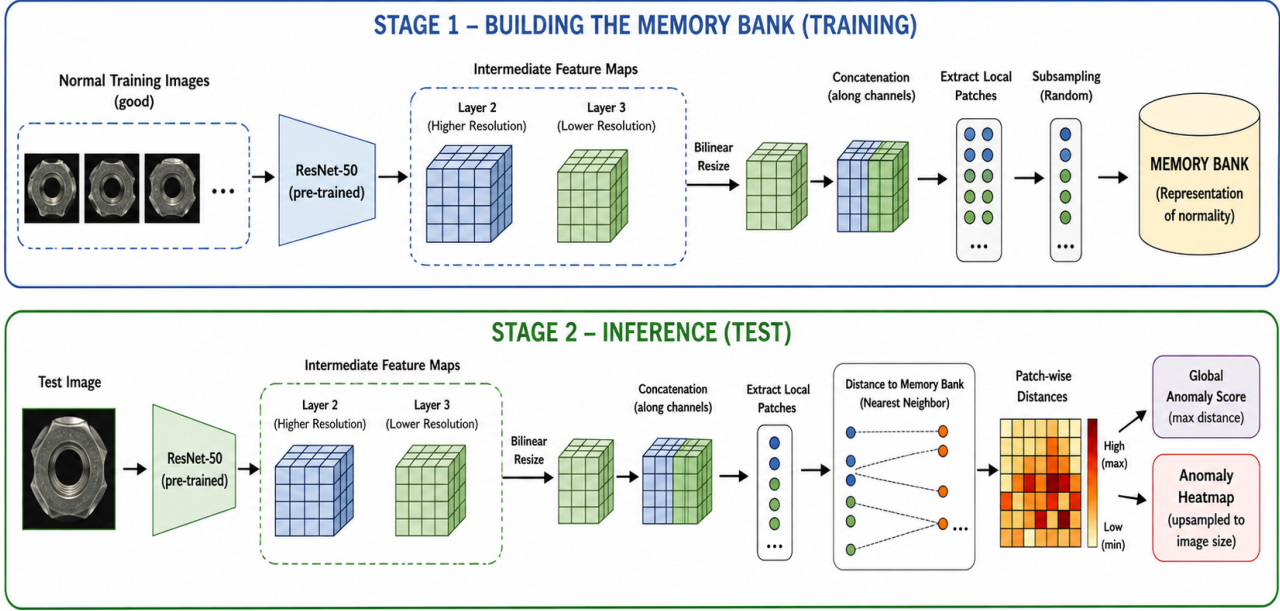
3.3. Anomaly Score

During inference, a test image is processed using the same ResNet-50 feature extraction pipeline. After resizing and concatenating the feature maps from *layer2* and *layer3*, the resulting representation is converted into a set of test patches

$$\mathcal{P} = \{p_1, p_2, \dots, p_K\}. \quad (3)$$

For each test patch p_i , the Euclidean distance from all elements of the memory bank is computed. The distance from the closest normal feature is then selected:

$$d_i = \min_{m_j \in \mathcal{M}} \|p_i - m_j\|_2. \quad (4)$$



Overall pipeline of the proposed method. Stage 1 builds the memory bank using only normal training images. Stage 2 computes patch-wise distances for a test image to obtain the global anomaly score and the anomaly heatmap.

A low value of d_i indicates that the patch is similar to a region observed in the normal training images, while a high value indicates a stronger deviation from the learned normal representation.

The global anomaly score assigned to the entire image is obtained by selecting the maximum patch distance:

$$S(x) = \max_i d_i. \quad (5)$$

The final classification threshold is estimated using the normal validation set. After constructing the memory bank, an anomaly score is computed for each validation image. The threshold T is then defined as the 90th percentile of the resulting score distribution:

$$T = P_{90}(\{S(x_v)\}_{x_v \in \mathcal{V}}), \quad (6)$$

where $\mathcal{V}$ denotes the validation set. A test image is classified as anomalous when $S(x) > T$.

3.4. Anomaly Localization

The same patch distances used for image-level classification are also exploited to provide an approximate localization of the defect. Since every distance d_i corresponds to a spatial location in the intermediate feature map, the values can be rearranged into a two-dimensional anomaly map.

The resulting map is normalized and resized to match the spatial resolution of the input image. A color map is then applied and superimposed on the original image, producing

a heatmap in which regions associated with larger feature distances are visually highlighted.

This representation provides additional information compared with the global anomaly score, since it allows the system to indicate which areas of the image contributed most strongly to the anomaly prediction.

4. Implementation

The proposed system was implemented in Python using PyTorch as the main deep learning framework. The implementation was divided into separate modules responsible for dataset management, feature extraction, anomaly detection, evaluation, and visualization. This modular organization simplifies the execution of the different stages of the pipeline and allows each component to be modified independently.

The main external libraries used in the implementation include *torch* and *torchvision* for neural network operations, PIL for image loading, NumPy for numerical operations, Scikit-learn for auxiliary feature processing, OpenCV for image resizing, and Matplotlib for the visualization and generation of anomaly heatmaps.

4.1. Model and Feature Extraction

The feature extractor is implemented as a custom PyTorch module containing a ResNet-50 model initialized with ImageNet pre-trained weights. All network parameters are frozen by setting their gradient computation to disabled,

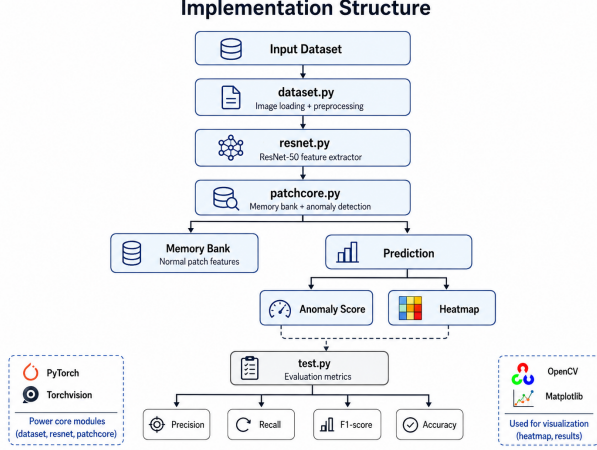


Figure 3: Software organization of the implemented anomaly detection system. The dataset module handles image loading and preprocessing, the ResNet-50 module extracts intermediate features, the PatchCore-based module constructs the memory bank and performs anomaly detection, while the evaluation module computes the final performance metrics.

since no fine-tuning is performed.

Forward hooks are registered on *layer2* and *layer3* in order to store their outputs during a forward pass. The final classification output of ResNet-50 is therefore ignored, while the intermediate feature maps are returned and subsequently processed by the anomaly detection module.

The PatchCore-based component receives the feature extractor and the execution device during initialization. During memory bank construction, batches of normal images are processed without gradient computation. The extracted feature maps are transferred to the CPU, resized when necessary, concatenated, and reshaped into local feature vectors.

The implementation also provides a configurable subsampling operation to reduce the number of vectors stored in the memory bank. A random subset of the available patch representations is retained according to a predefined percentage.

4.2. Inference and Classification

During inference, each test image is processed using the same preprocessing pipeline adopted for the training samples. An additional batch dimension is introduced before the image is provided to the feature extractor.

The extracted test patches are compared with the memory bank using the `torch.cdist` function, which computes pairwise Euclidean distances. For each test patch, only the minimum distance from the normal reference features is retained. The maximum value among these mini-

mum distances represents the global anomaly score of the image.

The classification threshold is not manually fixed. After the memory bank has been constructed, the 44 normal validation images are independently processed by the model and their anomaly scores are collected. The final threshold is selected as the 90th percentile of this distribution.

This procedure provides a threshold derived exclusively from normal samples that were not used for memory bank construction. Once determined, the threshold is kept fixed during the evaluation of both the original test set and the domain shift experiments.

An image is classified as anomalous when its anomaly score is greater than T ; otherwise, it is classified as normal.

The main implementation parameters are summarized in Table 3.

Table 3: Main parameters of the implemented anomaly detection system.

Parameter	Value
Backbone	ResNet-50
Pre-trained weights	ImageNet
Input resolution	224×224
Extracted layers	Layer 2, Layer 3
Feature interpolation	Bilinear
Distance metric	Euclidean
Train / validation split	80% / 20%
Memory bank reduction	Random subsampling (20%)
Threshold selection	Validation 90th percentile

4.3. Evaluation

The evaluation procedure iterates over all categories contained in the test set. Samples belonging to the *good* category are assigned the normal label, while all defect categories are considered anomalous.

For every test image, the predicted class is compared with the corresponding ground-truth label. The resulting predictions are used to compute the number of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN).

Based on these values, the performance of the system is evaluated using Accuracy, Precision, Recall, and F1-score.

Accuracy measures the overall proportion of correctly classified samples:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}. \quad (7)$$

Precision measures the proportion of samples classified as anomalous that are actually defective:

$$Precision = \frac{TP}{TP + FP}. \quad (8)$$

Recall measures the proportion of defective samples that are correctly identified by the system:

$$Recall = \frac{TP}{TP + FN}. \quad (9)$$

Finally, the F1-score represents the harmonic mean between Precision and Recall:

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}. \quad (10)$$

Accuracy, Precision, Recall, and F1-score together provide a complete picture of classification performance, capturing both overall correctness and the trade-off between false positives and false negatives.

The same threshold obtained from the normal validation set is used for all test samples. No information from the test labels is used during threshold selection.

4.4. Heatmap Generation

For anomalous test images, the system can additionally generate a visual representation of the localization result. The patch-level anomaly distances are first reshaped into a two-dimensional map and normalized between 0 and 1.

The anomaly map is then resized to the same spatial dimensions as the processed input image using OpenCV. A color map is applied to the normalized values and the resulting heatmap is superimposed on the original image.

In the implemented visualization, the final output is obtained by combining 60% of the original image with 40% of the colored anomaly map. The resulting figures contain both the original sample and its corresponding anomaly visualization, together with the global anomaly score computed by the model.

5. Results

The proposed anomaly detection system was evaluated on the test portion of the *metal_nut* category. Before testing, the 44 normal validation images were processed to estimate the classification threshold. The threshold was selected as the 90th percentile of the validation anomaly score distribution. The resulting threshold was then kept fixed for the complete test set. The quantitative evaluation is based on Accuracy, Precision, Recall, and F1-score, while the generated heatmaps are used for qualitative localization analysis.

The resulting threshold was $T = 36.02$.

5.1. Overall Classification Performance

The overall classification results obtained on the complete test set are reported in Table 4.

Table 4: Overall classification performance on the *metal_nut* test set.

Metric	Value
Accuracy	91.3%
Precision	96.6%
Recall	92.5%
F1-score	94.5%

The high Precision indicates that most samples classified as anomalous were actually defective. At the same time, the Recall above 92% shows that the system successfully detected the large majority of defective samples. The resulting F1-score confirms a good balance between false alarms and missed defects.

5.2. Confusion Matrix

A more detailed representation of the classification results can be obtained through the confusion matrix. Table 5 reports the number of True Positives, False Positives, True Negatives, and False Negatives produced by the system.

Table 5: Confusion matrix of the anomaly detection system.

	Predicted Normal	Predicted Anomaly
Actual Normal	TN = 19	FP = 3
Actual Anomaly	FN = 7	TP = 86

The confusion matrix shows that 19 of the 22 normal test images were correctly classified, while 3 normal samples were incorrectly identified as anomalous. Among the 93 defective images, 86 were correctly detected and 7 were incorrectly classified as normal.

The relatively small number of False Positives is consistent with the high Precision obtained by the system. False Negatives are particularly relevant in an industrial quality-control scenario, since they correspond to defective products that would incorrectly be considered acceptable. Nevertheless, the model successfully detected more than 92% of the anomalous samples.

5.3. Qualitative Localization Results

In addition to the numerical evaluation, the anomaly localization capability of the system was analyzed through the generated heatmaps. Some representative examples are shown in Fig. 4.

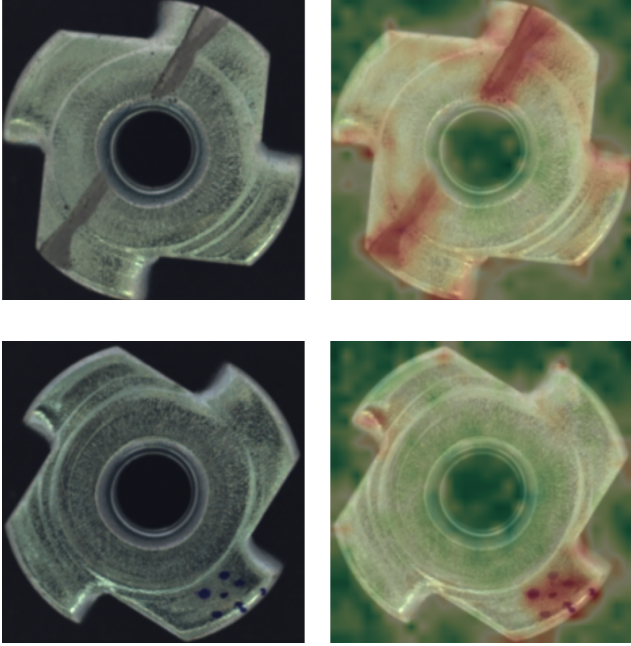


Figure 4: Examples of qualitative anomaly localization results. For each sample, the original image is compared with the corresponding heatmap generated from the patch-level anomaly distances. Regions associated with higher anomaly values are highlighted more strongly.

The qualitative results make it possible to verify whether the regions associated with high anomaly scores correspond to the actual defective areas. A successful localization is obtained when the highest values of the heatmap are concentrated around the visible defect, while low anomaly values are assigned to the remaining parts of the object.

In the selected examples, the regions associated with the highest anomaly values are concentrated around the visible defects, while most of the defect-free surface presents lower anomaly responses. This indicates that the patch-level distances used for classification also preserve useful spatial information for approximate defect localization.

5.4. Anomaly Score Analysis

The anomaly scores obtained on the test set show a clear difference between normal and defective samples. Normal images obtained an average score of 33.70, while anomalous images reached an average value of 43.00.

Table 6: Summary of the anomaly scores obtained on the test set.

Class	Mean Score	Std. Dev.
Normal	33.70	2.58
Anomalous	43.00	4.14

The difference between the two mean scores indicates that the feature representation stored in the memory bank provides a useful separation between normal and anomalous samples. The selected threshold of $T = 36.02$ lies between the average scores of the two classes and allows most defective images to be detected while limiting the number of False Positives.

However, the non-zero classification errors indicate that the two score distributions are not completely separated. Some defective samples produce scores similar to normal images, while a small number of normal samples exceed the selected threshold.

6. Domain Shift

In real industrial environments, acquisition conditions may differ from those used to construct the normal memory bank. Variations in illumination, sensor response, image quality, or camera viewpoint can modify the extracted features even when no actual defect is present. To evaluate the robustness of the proposed system, several perturbations were therefore applied to the original test images while preserving their ground-truth labels.[5]

6.1. Perturbations and Evaluation Protocol

Five transformations were selected for the robustness evaluation: *exposure*, *white balance*, *noise*, *contrast*, and *perspective*.

Table 7: Perturbation models used for domain shift evaluation.

Transform	Model	Range
Exposure	$I' = \alpha I + \beta$	$\alpha \in [0.5, 1.7]$
White balance	per-channel scaling (R, B)	$[0.70, 1.40]$
Noise	Gaussian + salt-and-pepper	$\sigma \in [5, 40]$
Contrast	CLAHE on L channel (LAB space)	clip $\in [2, 6]$
Perspective	homography from pitch/roll tilt	$\pm 15^\circ$

Exposure reproduces incorrect shutter timing, lamp aging, and voltage fluctuations in industrial lighting, while white balance reproduces the switch between different lighting technologies (fluorescent, LED, halogen). Noise models sensor read noise and dead/hot pixels at varying gain. Contrast reproduces AGC and tone-curve differences between camera units, and Perspective models a camera remounted with a slight tilt after maintenance.

In addition to these five, other transforms inspired by realistic industrial scenarios were also implemented but not included in the final evaluation. JPEG compression reproduces bandwidth-limited camera links or on-device compression, gamma correction reproduces a misconfigured or replaced camera with a different tone curve, blur (motion and defocus) reproduces vibration or object height variation on the conveyor, vignetting reproduces uneven lens illumination toward the image borders, and specular highlights

Table 8: Classification performance under domain shift.

Condition	Acc.	Prec.	Rec.	F1
Original	91.3%	96.6%	92.5%	94.5%
Exposure	89.0%	91.6%	95.0%	93.3%
White Balance	89.0%	89.6%	97.5%	93.4%
Noise	90.0%	91.6%	96.3%	93.9%
Contrast	85.0%	84.2%	100%	91.4%
Perspective	85.0%	87.4%	95.0%	91.0%

reproduce glare on glossy metal surfaces. These were excluded from the final evaluation since the selected five already cover the main sources of domain shift relevant to this scenario.

Representative examples are shown in Fig. 5.

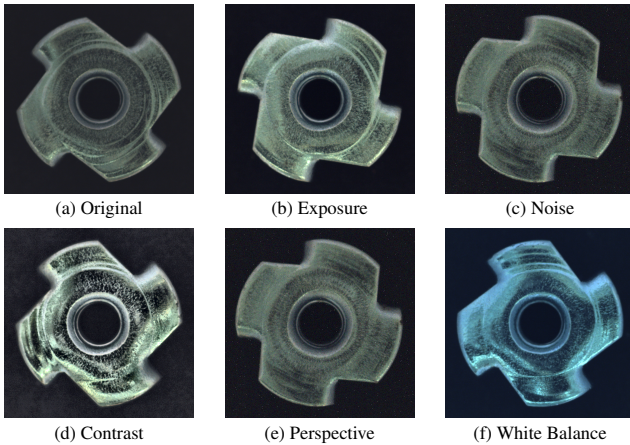


Figure 5: Representative domain shift transformations applied to the test images.

For each transformation, the parameters are randomly sampled within predefined ranges. The shifted images are used exclusively during evaluation: the ResNet-50 feature extractor, memory bank, and anomaly threshold remain unchanged, and no additional training or calibration is performed. Accuracy, Precision, Recall, and F1-score are computed for each condition and compared with the results obtained on the original test set.

6.2. Results and Discussion

The robustness results are summarized in Table 8.

The comparison with the nominal results makes it possible to identify which acquisition variations have the strongest impact on the anomaly detector. Particular attention is given to Recall, since a reduction in this metric corresponds to an increase in defective samples incorrectly classified as normal. The experiments also provide an indication of whether the fixed anomaly threshold remains reliable when illumination, sensor characteristics, or camera viewpoint differ from the nominal acquisition conditions.

7. Conclusion

This work presented an anomaly detection system for the identification and localization of defects in industrial images. The proposed approach combines a pre-trained ResNet-50 feature extractor with a PatchCore-based nearest-neighbor strategy. Defect-free training images are used to construct a memory bank of local features, while test samples are classified according to their distance from the stored normal representations.

The system was evaluated on the *metal_nut* category of the MVTec AD dataset using Accuracy, Precision, Recall, and F1-score. Patch-level anomaly distances were also used to generate heatmaps for approximate defect localization.

Robustness under domain shift was evaluated using exposure changes, white-balance variations, sensor noise, contrast modifications, and perspective transformations, simulating variations in illumination, acquisition quality, and camera viewpoint without introducing actual defects.

Under nominal conditions, the system achieved an Accuracy of 91.%, a Precision of 96.6%, a Recall of 92.5%, and an F1-score of 94.5%. The confusion matrix showed 86 correctly detected anomalies, with only 3 False Positives and 7 False Negatives.

Overall, the results show that the proposed approach can effectively distinguish normal and defective samples while providing useful spatial information for approximate anomaly localization.

References

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